

< Homework #5 >

1. i)
$$\begin{bmatrix} I_x \\ I_y \\ I_z \end{bmatrix} = \begin{bmatrix} 1.4\omega \\ 1.7\omega \\ 1.\omega \end{bmatrix} \text{ kg-m}^2$$

ii)
$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{t=0} = \begin{bmatrix} 0.5 \\ 0.3 \\ 0 \end{bmatrix} \text{ rad}, \quad \begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{t=t_f} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

iii) $t_f = 120 \text{ sec}$

iv) $h = 300 \text{ km}$

$$R = R_E + h = 6,700 \text{ km}$$

$$\omega_0 = \sqrt{\mu/R^3} = 1.1512 \times 10^{-3} \text{ rad/sec}$$

v) On-orbit rotational EOM including gravity gradient

vi) Slewing maneuver about a constant axis w.r.t. LVLH frame

a) Since $R_{BL} = R_x(\phi) R_y(\theta) R_z(\psi)$,

$$R_0 = R(t=0)$$

$$= R_{LB}(t=0)$$

$$= R_z(-\psi_0) R_y(-\theta_0) R_x(-\phi_0)$$

$$= \begin{bmatrix} 0.8384 & -0.4794 & 0.2593 \\ 0.4580 & 0.8776 & 0.1417 \\ -0.2955 & 0 & 0.9553 \end{bmatrix}$$

Converting to exponential coordinates

$$\bar{\Phi}_0 = \bar{\Phi}(t=0)$$

$$= \cos^{-1} \left(\frac{\text{tr}(R_0) - 1}{2} \right)$$

$$= 0.5815 \text{ rad} = 33.3161 \text{ deg.}$$

(3)

$$a = \frac{1}{2 \sin \Phi_0} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \begin{bmatrix} -0.1290 \\ 0.5051 \\ 0.8534 \end{bmatrix}$$

Simply from $R_f = R(t=t_f) = I_{3 \times 3}$, which implies no rotation about any axis,

$$\Phi_f = \Phi(t=t_f) = 0.$$

Summarized

$$\left(\begin{array}{l} a = \begin{bmatrix} -0.1290 \\ 0.5051 \\ 0.8534 \end{bmatrix} \\ \Phi_0 = 0.5815 \text{ rad} = 33.3161 \text{ deg} \\ \Phi_f = 0 \end{array} \right.$$

b) Consider the rest-to-rest slewing maneuver w.r.t. LVLH frame about the constant 'a' - axis

$$\omega^r(t) = a \dot{\Phi}(t)$$

$$\Phi(t=0) = 0.5815 \text{ rad} \quad \Phi(t=t_f) = 0 \Leftarrow \text{given from (a)}$$

$$\dot{\Phi}(t=0) = 0 \quad \dot{\Phi}(t=t_f) = 0 \Leftarrow \text{rest-to-rest.}$$

Using the form

$$\Phi(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3$$

It turns out that

$$\Phi(t) = \Phi_0 - 3\Phi_0 \left(\frac{t}{t_f}\right)^2 + 2\Phi_0 \left(\frac{t}{t_f}\right)^3$$

Now, from the relation,

$$\vec{\omega}^r = \vec{\omega} - R_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix}$$

$$\begin{aligned}
 \vec{\omega}(t) &= \vec{\omega}^r(t) + R_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix} \\
 &= a \ddot{\Phi}(t) + R_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix} \\
 &= a(b_1 + 2b_2 t + 3b_3 t^2) + R_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \dot{\vec{\omega}}(t) &= a \ddot{\dot{\Phi}}(t) + \dot{R}_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix} \\
 &= a(2b_2 + 6b_3 t) + S(\vec{\omega}) R_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix} \\
 &= a(2b_2 + 6b_3 t) + S(a \dot{\Phi}) R_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix}
 \end{aligned}$$

Finally, the actuator torque M_A can be evaluated ~~by~~ as

$$M_A = I\dot{\omega} + \omega \times I\omega - M_G$$

where M_G is the gravity gradient torque.

The following plots justify this approach and show that all the boundary conditions are satisfied.

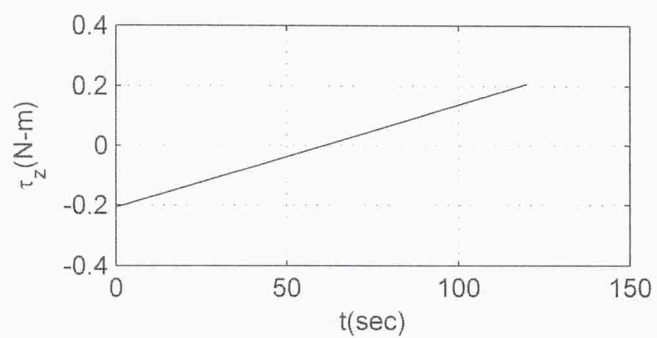
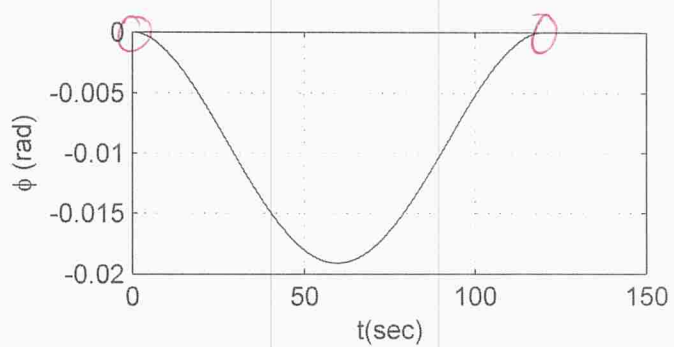
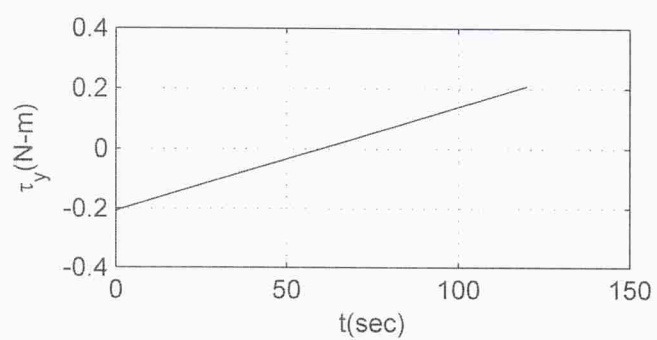
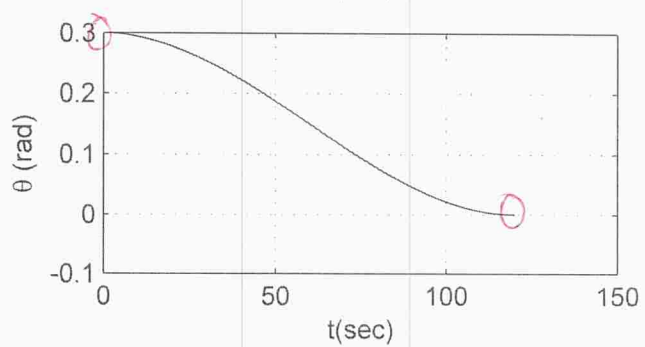
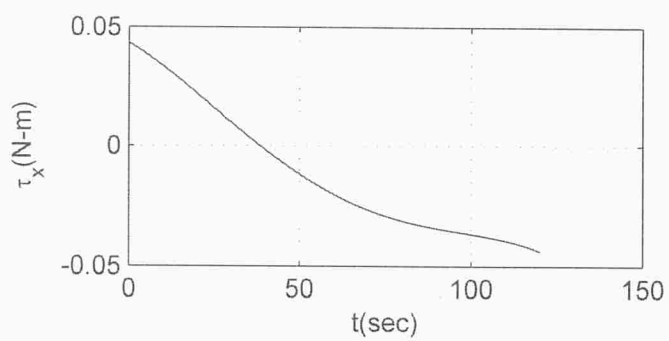
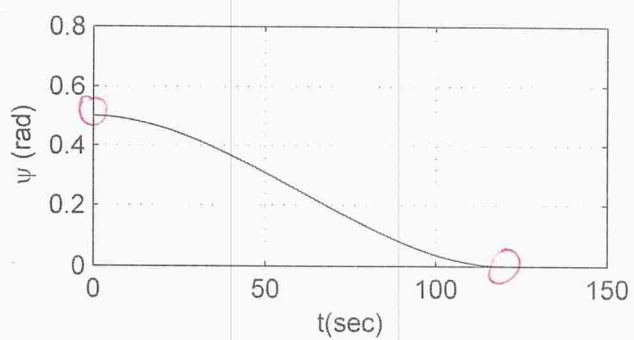
That is,

$$\omega(t_f) = \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix}$$

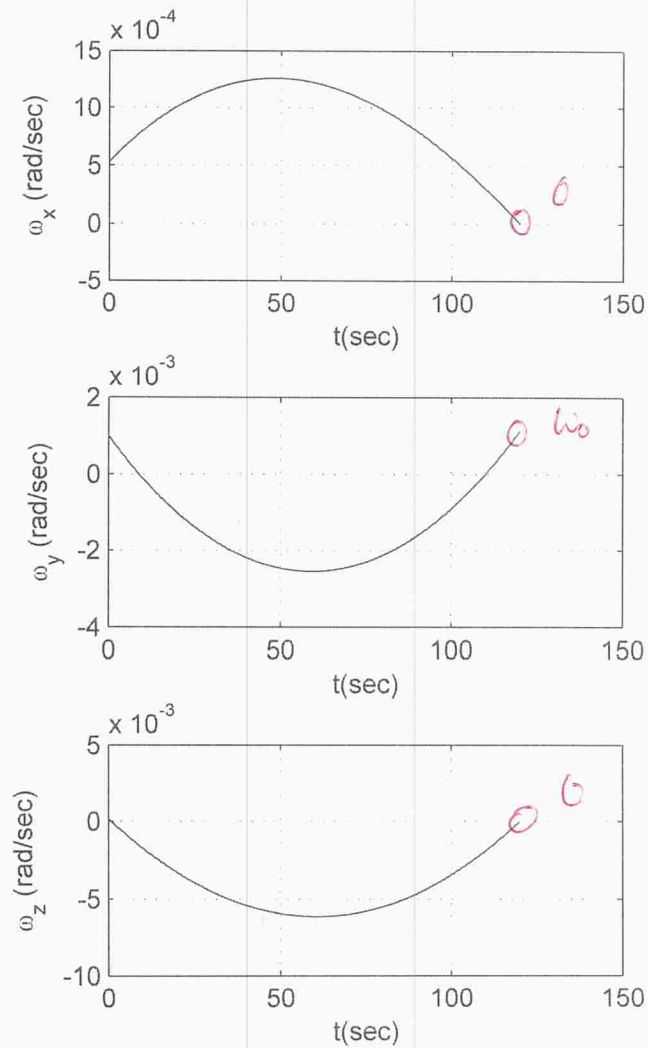
$$\omega^r(t_0) = \omega^r(t_f) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{t=0} = \begin{bmatrix} 0.5 \\ 0.3 \\ 0 \end{bmatrix}$$

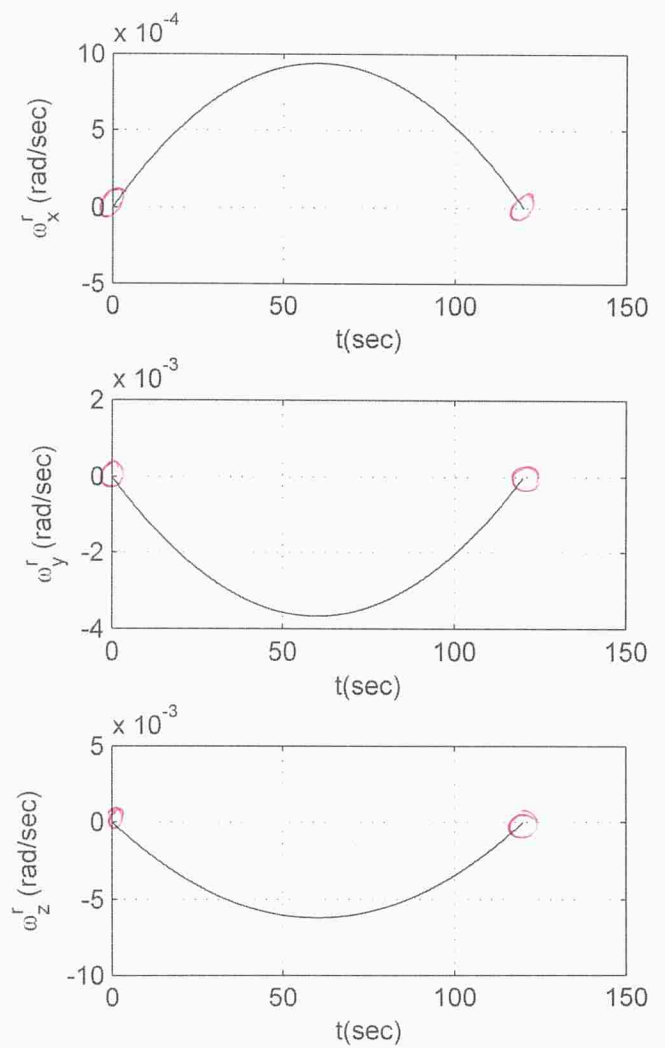
$$\begin{bmatrix} \psi \\ \theta \\ \phi \end{bmatrix}_{t=f} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



Angular Velocity
w.r.t. Inertial Frame



Angular Velocity
w.r.t. LVLH frame



2. Linearized EoM for pitch equation is

$$\begin{bmatrix} \dot{\Delta \theta} \\ \Delta \dot{w}_y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -9.3550 \times 10^{-7} & 0 \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta w_y \end{bmatrix} + \begin{bmatrix} 0 \\ 5.8824 \times 10^{-4} \end{bmatrix} M_y$$

Linearized EoM for roll and yaw motion is

$$\begin{bmatrix} \dot{\Delta \phi} \\ \Delta \dot{w}_x \\ \dot{\Delta \psi} \\ \Delta \dot{w}_z \end{bmatrix} = \begin{bmatrix} 0 & 1 & -1.1512 \times 10^{-3} & 0 \\ -1.9879 \times 10^{-4} & 0 & 0 & 5.7561 \times 10^{-4} \\ 1.1512 \times 10^{-3} & 0 & 0 & 1 \\ 0 & -3.4536 \times 10^{-4} & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \phi \\ \Delta w_x \\ \Delta \psi \\ \Delta w_z \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 7.1449 \times 10^{-4} & 0 \\ 0 & 0 \\ 0 & 1 \times 10^{-3} \end{bmatrix} \begin{bmatrix} M_x \\ M_z \end{bmatrix}$$

Boundary Conditions to be satisfied are

$$\begin{bmatrix} \Delta \psi \\ \Delta \theta \\ \Delta \phi \end{bmatrix}_{t=t_0} = \begin{bmatrix} 0.5 \\ 0.3 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \Delta \psi \\ \Delta \theta \\ \Delta \phi \end{bmatrix}_{t=t_f} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \Delta w_x \\ \Delta w_y \\ \Delta w_z \end{bmatrix}_{t=0} = R_0 \begin{bmatrix} 0 \\ w_0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ w_0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5.2727 \times 10^{-4} \\ -1.4903 \times 10^{-4} \\ +1.6310 \times 10^{-4} \end{bmatrix}$$

$$\begin{bmatrix} \Delta w_x \\ \Delta w_y \\ \Delta w_z \end{bmatrix}_{t=t_f} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

a) Let $\Delta T_y(0)$ and $\Delta T_y(t_f^-)$ be the initial and final impulse, respectively. Then, from

$$\begin{bmatrix} \Delta \theta(t_f) \\ \Delta \omega_y(t_f^-) \end{bmatrix} = \begin{bmatrix} 0 \\ \Delta \omega_y(t_f^-) \end{bmatrix} = e^{At_f} \begin{bmatrix} 0.3 \\ \Delta \omega_y(0^+) \end{bmatrix} = e^{At_f} \begin{bmatrix} \Delta \theta(0) \\ \Delta \omega_y(0^+) \end{bmatrix}$$

$$\begin{cases} \Delta T_y(0) = \Delta \omega_y(0^+) - \Delta \omega_y(0) = -2.3478 \times 10^{-3} \text{ rad/sec} \\ \Delta T_y(t_f^-) = \Delta \omega_y(t_f) - \Delta \omega_y(t_f^-) = 2.5056 \times 10^{-3} \text{ rad/sec} \end{cases}$$

b) Similarly from

$$\begin{bmatrix} \Delta \phi(t_f) \\ \Delta \omega_x(t_f) \\ \Delta \psi(t_f) \\ \Delta \omega_z(t_f^-) \end{bmatrix} = \begin{bmatrix} 0 \\ \Delta \omega_x(t_f^-) \\ 0 \\ \Delta \omega_z(t_f^-) \end{bmatrix} = e^{At_f} \begin{bmatrix} 0 \\ \Delta \omega_x(0^+) \\ 0.5 \\ \Delta \omega_z(0^+) \end{bmatrix} = e^{At_f} \begin{bmatrix} \Delta \phi(0) \\ \Delta \omega_x(0^+) \\ \Delta \psi(0) \\ \Delta \omega_z(0^+) \end{bmatrix}$$

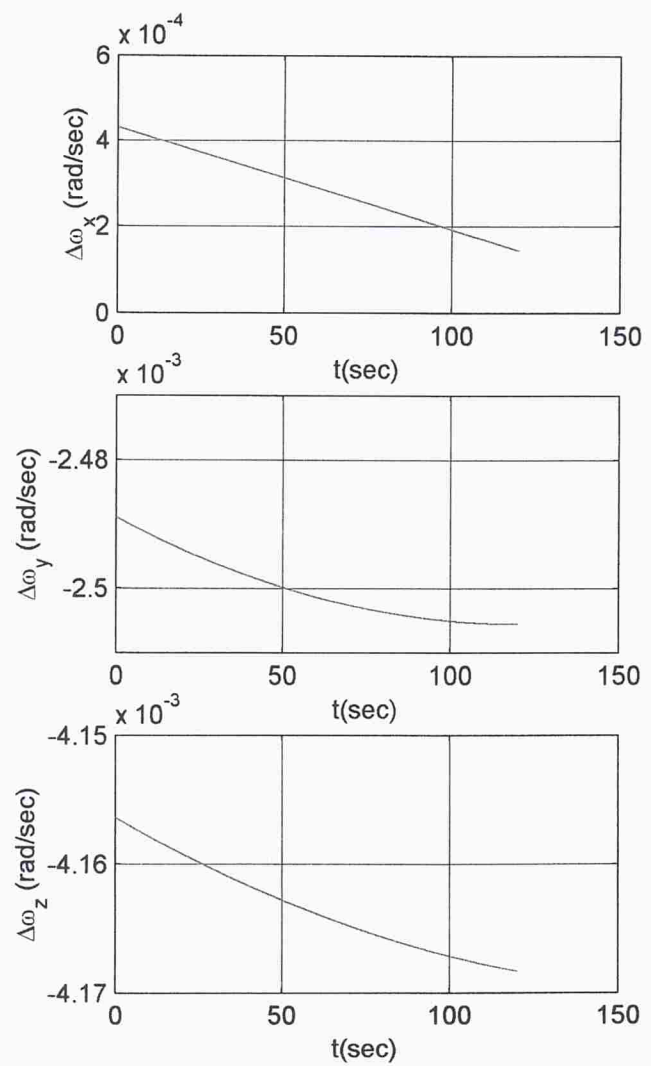
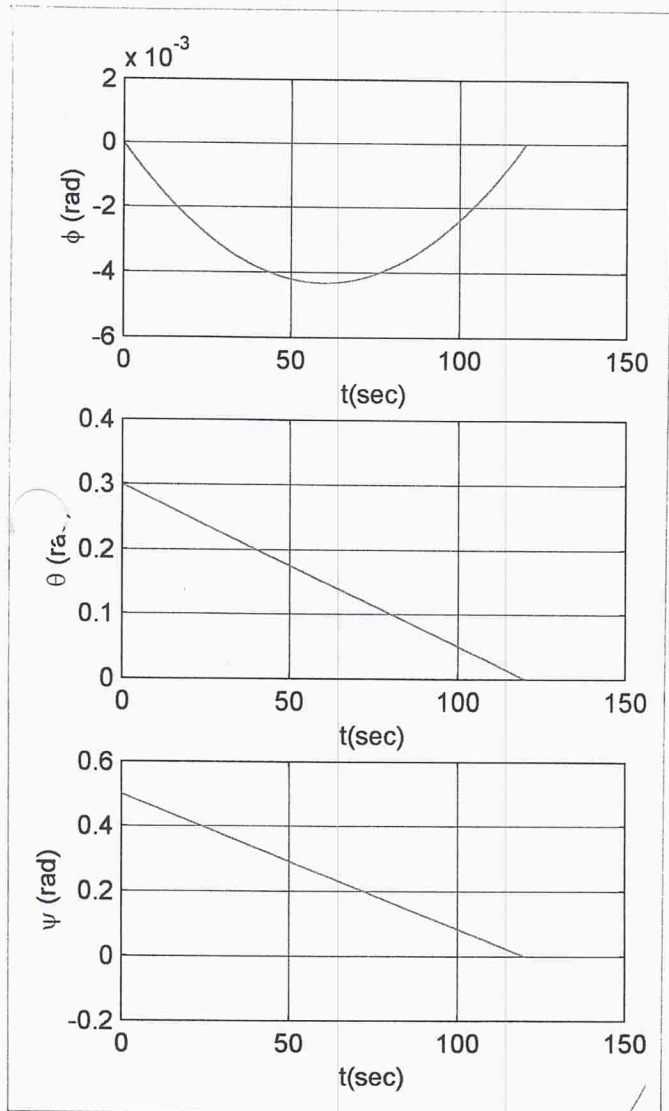
$$\begin{cases} \Delta T_x(0) = \Delta \omega_x(0^+) - \Delta \omega_x(0) = \begin{matrix} -1.4422 \times 10^{-4} \text{ rad/sec} \\ \cancel{-9.5956 \times 10^{-5} \text{ rad/sec}} \end{matrix} \\ \Delta T_x(t_f^-) = \Delta \omega_x(t_f) - \Delta \omega_x(t_f^-) = -1.4443 \times 10^{-4} \text{ rad/sec} \end{cases}$$

$$\begin{cases} \Delta T_z(0) = \Delta \omega_z(0^+) - \Delta \omega_z(0) = -4.3195 \times 10^{-3} \text{ rad/sec} \\ \cancel{\Delta T_z(0)} \\ \Delta T_z(t_f^-) = \Delta \omega_z(t_f) - \Delta \omega_z(t_f^-) = 4.1683 \times 10^{-3} \text{ rad/sec} \end{cases}$$

Finally the following plots show the attitude responses

Prob 2 (a), (b)

8



(9)

$$c) \begin{bmatrix} \Delta \theta(t_f) \\ \Delta w_y(t_f) \end{bmatrix} = e^{At_f} \begin{bmatrix} \Delta \theta(0) \\ \Delta w_y(0) \end{bmatrix} + \int_0^{t_s} e^{A(t_f-s)} B \bar{M}_y ds + \int_{t_s}^{t_f} e^{A(t_f-s)} B (-\bar{M}_y) ds$$

$$\text{where } \begin{bmatrix} \Delta \theta(0) \\ \Delta w_y(0) \end{bmatrix} = \begin{bmatrix} 0.3 \\ 1.4903 \times 10^{-4} \end{bmatrix}, \quad \begin{bmatrix} \Delta \theta(t_f) \\ \Delta w_y(t_f) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving iteratively for t_s and $\bar{M}_y$

$$\begin{bmatrix} t_s \\ \bar{M}_y \end{bmatrix} = \begin{bmatrix} 59.069 \text{ sec} \\ -1.3730 \times 10^{-1} \text{ N-m} \end{bmatrix}$$

d) Similarly solving iteratively for $(t_s^x, t_s^z, \bar{M}_x, \bar{M}_z)$ with

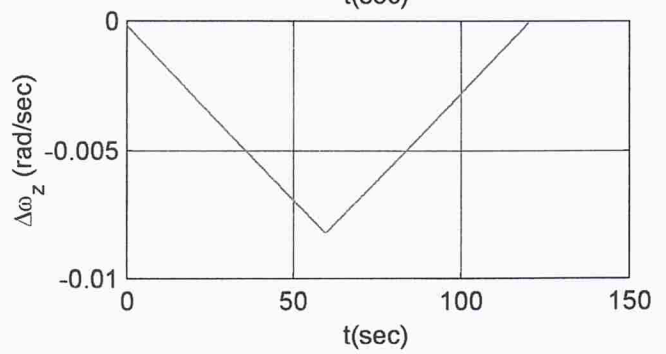
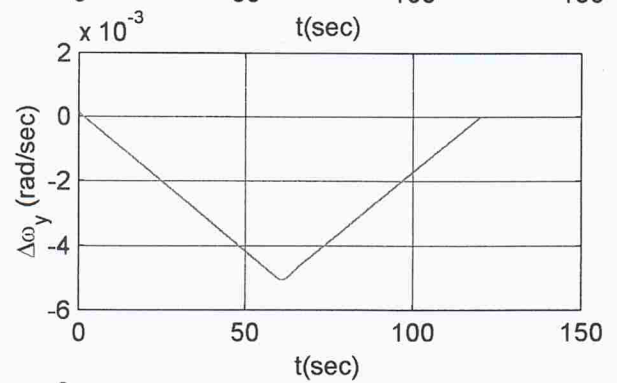
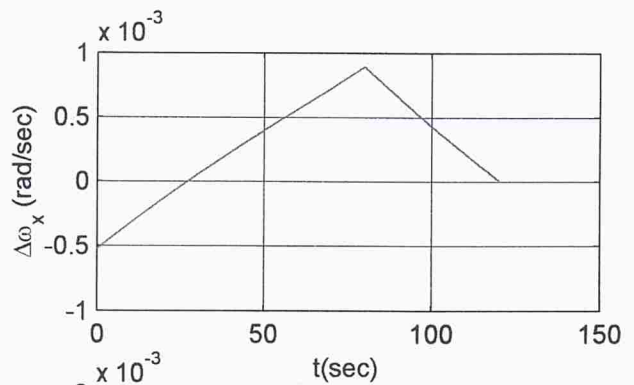
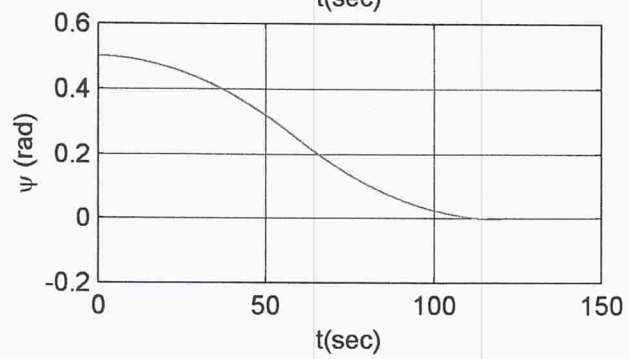
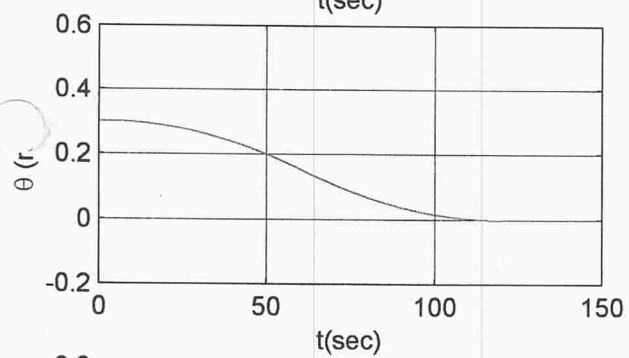
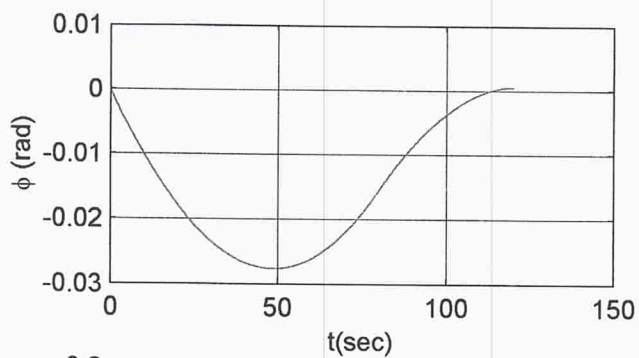
$$\begin{bmatrix} \Delta \phi(0) \\ \Delta w_x(0) \\ \Delta \psi(0) \\ \Delta w_z(0) \end{bmatrix} = \begin{bmatrix} 0 \\ -5.2727 \times 10^{-4} \\ 0.5 \\ -1.6310 \times 10^{-4} \end{bmatrix}, \quad \begin{bmatrix} \Delta \phi(t_f) \\ \Delta w_x(t_f) \\ \Delta \psi(t_f) \\ \Delta w_z(t_f) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} t_s^x \\ \bar{M}_x \end{bmatrix} = \begin{bmatrix} 79.725 \text{ sec} \\ 2.8752 \times 10^{-2} \text{ N-m} \end{bmatrix}$$

$$\begin{bmatrix} t_s^z \\ \bar{M}_z \end{bmatrix} = \begin{bmatrix} 59.355 \text{ sec} \\ -1.3569 \times 10^{-1} \text{ N-m} \end{bmatrix}$$

Finally, the following plots show the response for the attitude angles (ψ, θ, ϕ)

10



Prob3. ① Nonlinear Control Approach

- Global behavior of the system can be observed
- It is difficult to formulate properly due to nonlinearity. The constant axis rotation is rather simple in formulation. However, it is confined to the maneuvering w.r.t. constant axis only. And the general multi-axis maneuver is difficult to formulate with nonlinear system ✓

② Two vector impulse with linearized system

- The concept is simple and easy to manipulate even for multi-axis rotation. ✓
- Due to the elimination of nonlinearity, only the region near the operating point can be observed accurately. ✓

③ Piecewise Constant Approach ✓

- Has the same strength and weakness as the two impulse approach.